

Lab 11: Eigenvalues and Eigenvectors, Convexity

EECS 245, Winter 2026 at the University of Michigan

due by the end of your lab section

Name: _____

username: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: Eigenvalues and Eigenvectors

Let $A = \begin{bmatrix} 6 & 3 \\ 3 & -2 \end{bmatrix}$.

- An **eigenvector** of A is a non-zero vector $\vec{v}$ such that $A\vec{v} = \lambda\vec{v}$ for some scalar λ . The scalar λ is called the **eigenvalue** corresponding to $\vec{v}$. For A 's eigenvectors, multiplying by A is equivalent to multiplying by a scalar.
- The **characteristic polynomial** of A is given by $p(\lambda) = \det(A - \lambda I)$.

$$p(\lambda) = \det(A - \lambda I) = \begin{vmatrix} 6 - \lambda & 3 \\ 3 & -2 - \lambda \end{vmatrix} = (6 - \lambda)(-2 - \lambda) - 3 \cdot 3 = \lambda^2 - 4\lambda - 21 = (\lambda + 3)(\lambda - 7)$$

- The eigenvalues of A are the roots of the characteristic polynomial, so $\lambda_1 = -3$ and $\lambda_2 = 7$.
 - The eigenvector $\vec{v}_1$ satisfies $A\vec{v}_1 = -3\vec{v}_1$.

$$\begin{bmatrix} 6 & 3 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = -3 \begin{bmatrix} a \\ b \end{bmatrix} \implies b = -3a$$

So any vector of the form $\begin{bmatrix} a \\ -3a \end{bmatrix}$ ($a \neq 0$) is an eigenvector of A corresponding to the

eigenvalue -3 . We could pick $\vec{v}_1 = \begin{bmatrix} 2 \\ -6 \end{bmatrix}$.

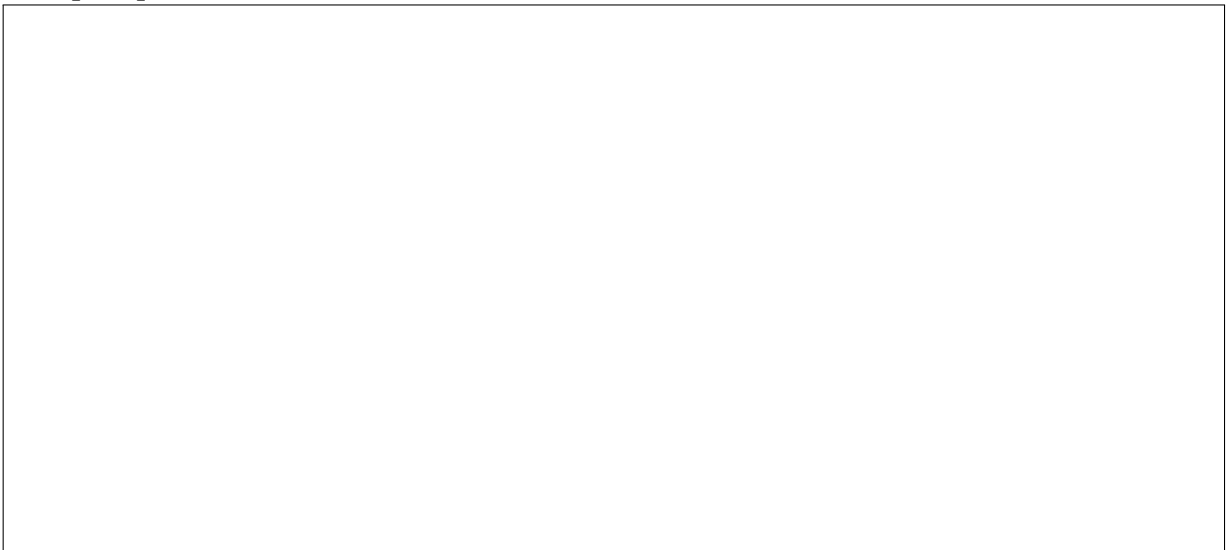
- The eigenvector $\vec{v}_2$ satisfies $A\vec{v}_2 = 7\vec{v}_2$. Another way to find it is to solve for the null space of $A - 7I = \begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix}$. One vector in $\text{nullsp}(A - 7I)$ is $\vec{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

Activity 1

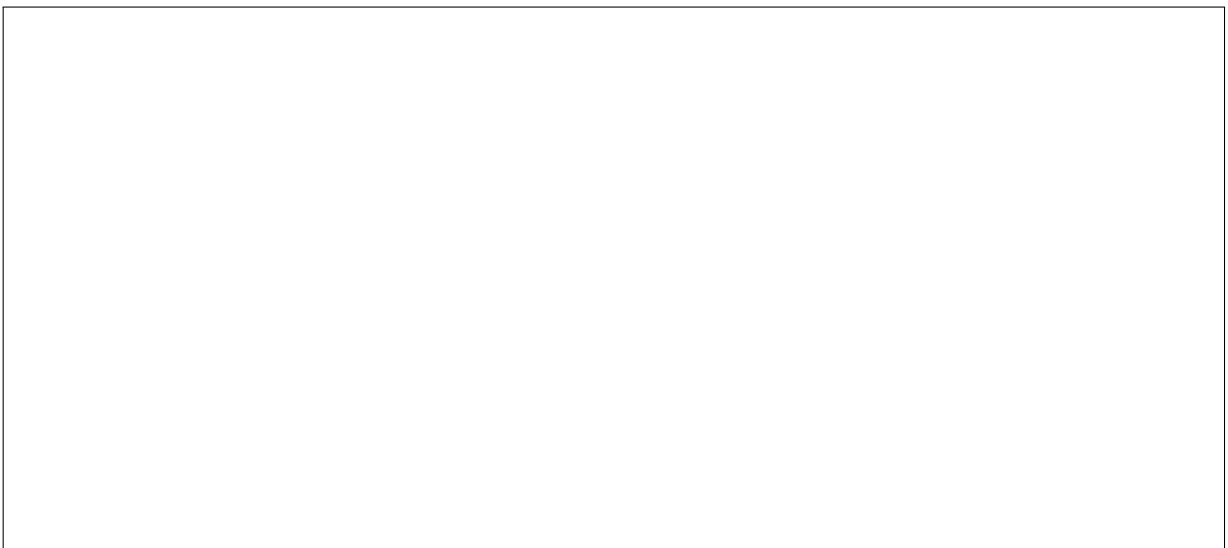
For each 2×2 matrix A below:

- (i) Find the characteristic polynomial of A , and use it to find the eigenvalues of A .
- (ii) Find one eigenvector for each eigenvalue of A . Verify that each eigenvector is indeed an eigenvector of A by multiplying it by A .
- (iii) **By hand (not using Python or Desmos)**, draw a picture (like the one in Chapter 9.1 titled [Visualizing the eigenvectors of \$A\$](#)) with vectors $\vec{v}_1, A\vec{v}_1, \vec{v}_2, A\vec{v}_2$ as arrows (where $\vec{v}_1$ and $\vec{v}_2$ are the eigenvectors you found above).

a) $A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$



b) $A = \begin{bmatrix} 3 & 4 \\ 4 & 3 \end{bmatrix}$



Activity 2: Rapid Fire

The goal of this activity is to practice spotting eigenvalues and characteristic polynomials quickly. Two quick facts:

- The **sum** of the eigenvalues of a matrix is equal to the **trace** of the matrix (which is the sum of the diagonal entries).
- The **product** of the eigenvalues of a matrix is equal to the **determinant** of the matrix.

a) A 2×2 matrix A has $\text{trace}(A) = 5$ and $\det(A) = 6$. What are the eigenvalues of A ?

b) A non-invertible 2×2 matrix has an eigenvalue of 5. What is its characteristic polynomial?

c) A 3×3 matrix A has $\det(A) = 20$ and two unique **positive integer** eigenvalues, one of which is repeated twice. In other words, $p(\lambda)$ has the form

$$p(\lambda) = (\lambda - \lambda_1)^2(\lambda - \lambda_2)$$

(λ_1 has an **algebraic multiplicity** of 2. This is a term we'll see more next week.)

What are all possible values of λ_1 and λ_2 ?

Activity 3: Quadratic Forms Return

Open Desmos in 3D mode at desmos.com/3d and write $z = x^2 + 2bxy + 16y^2$. This should show you a 3D surface along with a slider for b . Drag the slider to see how the shape of the surface changes for different b 's. You should notice that depending on the value of b , the surface may or may not have a global minimum. Let's explore!

- a) z is a quadratic form, $f(\vec{x}) = \vec{x}^T A \vec{x}$, where $\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}$ and A is a symmetric matrix. Find A .

- b) For a vector-to-scalar function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the **Hessian** of f , denoted $\nabla^2 f$, is the $n \times n$ matrix of second partial derivatives of f . Find $\nabla^2 f$ for $f(\vec{x}) = \vec{x}^T A \vec{x}$.

- c) A symmetric matrix A is **positive semidefinite** (PSD) if $\vec{v}^T A \vec{v} \geq 0$ for all $\vec{v} \in \mathbb{R}^n$. In English, this says that A is positive semidefinite if the quadratic form $f(\vec{v}) = \vec{v}^T A \vec{v}$ is always non-negative for all $\vec{v} \in \mathbb{R}^n$. Two relevant facts:

- A differentiable vector-to-scalar function f is **convex** if its Hessian is PSD.
- A symmetric matrix A is PSD if and only if all of its eigenvalues are non-negative.

Using the facts above, find the range of values b for which f is convex, and verify your answer by dragging the slider on Desmos.

Activity 4: Understanding Complex Proofs

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. It turns out that the function $g(\vec{x})$, defined by

$$g(\vec{x}) = f(A\vec{x} + \vec{b})$$

for some $n \times n$ matrix A and vector $\vec{b} \in \mathbb{R}^n$, is also convex, no matter what A and $\vec{b}$ are. We're not going to ask you to prove this on your own: instead, we'll give you a proof and ask you questions to ensure you understand it.

Our **goal** is to show that $g((1-t)\vec{x} + t\vec{y}) \leq (1-t)g(\vec{x}) + tg(\vec{y})$, for all $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $t \in [0, 1]$. We'll start with the "left-hand side" of the definition, and try and leverage f 's convexity.

$$g((1-t)\vec{x} + t\vec{y}) = f\left(A((1-t)\vec{x} + t\vec{y}) + \vec{b}\right) \quad (1)$$

$$= f\left((1-t)A\vec{x} + tA\vec{y} + \vec{b}\right) \quad (2)$$

$$= f\left((1-t)(A\vec{x} + \vec{b}) + t(A\vec{y} + \vec{b})\right) \quad (3)$$

$$\leq (1-t)f(A\vec{x} + \vec{b}) + tf(A\vec{y} + \vec{b}) \quad (4)$$

$$= \boxed{(1-t)g(\vec{x}) + tg(\vec{y})} \quad (5)$$

a) In which line did we use the fact that f is convex?

b) How did we move from line (1) to line (2), i.e. $f\left(A((1-t)\vec{x} + t\vec{y}) + \vec{b}\right) = f\left((1-t)A\vec{x} + tA\vec{y} + \vec{b}\right)$?

c) How did we move from line (2) to line (3), i.e. $f\left((1-t)A\vec{x} + tA\vec{y} + \vec{b}\right) = f\left((1-t)(A\vec{x} + \vec{b}) + t(A\vec{y} + \vec{b})\right)$?

Recall, $g(\vec{x}) = f(A\vec{x} + \vec{b})$, where A is an $n \times n$ matrix and $\vec{x}, \vec{b} \in \mathbb{R}^n$. On the last page, we showed that if f is convex, then g is convex.

Now, let's explore what happens if f is **strictly** convex. Recall, this means that for all (non-equal) $\vec{x}$ and $\vec{y}$ in its domain, and for any $t \in (0, 1)$,

$$f((1-t)\vec{x} + t\vec{y}) < (1-t)f(\vec{x}) + tf(\vec{y})$$

- d) Suppose $\text{rank}(A) = n$. Explain why it's impossible for $A\vec{x} + \vec{b} = A\vec{y} + \vec{b}$ for two different vectors $\vec{x}$ and $\vec{y}$.

- e) Suppose $\text{rank}(A) < n$. Explain why it's possible for $g(\vec{x}) = g(\vec{y})$ for two different vectors $\vec{x}$ and $\vec{y}$. *Hint: Think about $\text{nullsp}(A)$.*

- f) Using the above reasoning, explain why if f is strictly convex, then g is strictly convex if $\text{rank}(A) = n$, and is (not strictly) convex if $\text{rank}(A) < n$.

- g) What were your thoughts on this type of activity, where we give you a proof and ask you questions about it?

☐ Hated it ☐ Didn't like it ☐ Neutral ☐ Liked it ☐ Loved it